

A located map of the fungal polyketide problem

boundary, frontier, and coverage, characterized by a sound executor’s reach

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Abstract

Predicting a fungal natural product from its biosynthetic gene cluster (BGC) is unsolved, but “unsolved” has not been a precise statement. We characterize *where* the problem lies and *what kind* of hardness each piece is, over the MIBiG-deposited fungal PKS manifold (326 entries, biased toward well-studied compounds), by the reach of a sound inverse compiler: a deterministic executor renders a biosynthetic program to a structure, and a verifier returns the exact consistent set Z^* of programs an observation admits (Eilertsen, 2026). The map has three panels. **(i) Boundary:** exact reconstruction reaches 8/326—the boundary of the *current sound grammar’s reach*, not a biological limit on the manifold. **(ii) Frontier:** within the reachable cores, the conformational decision frontier is thin and localized, and exact by complete enumeration—zero over the non-methylating aromatic interiors, single-digit within the highly-reducing subclass. **(iii) Coverage census:** the out-of-grammar remainder partitions into an *in-scope* tailoring gap (CHO chemistry a polyketide grammar could in principle reach), a *heteroatom scope boundary* (a different grammar’s problem), and a small inconclusive bucket; a re-scan with a lifted size cap establishes that the boundary is genuine rather than a tractability artifact—0 of 91 conclusively tested large cores were reachable. A structural-proxy set-cover analysis—an explicit upper bound—shows the in-scope hardness is concentrated in a few coarse families but dominated by non-aromatic ring/macrolactonization and overwhelmingly entangled: 86% of in-scope cores touch a family whose sound expressibility is unproven, and the soundly-extendable part—what a soundness-preserving build with no new regiochemistry recovers—is $\approx$ one core. The contribution is the located map itself: it partitions the unsolved into *buildable*, *research-bet*, and *different-grammar*, telling the field where to spend effort and what kind each piece needs—a more useful object, we argue, than a partial solver.

1 The question: where does a sound executor’s reach stop?

A product is the image $y = \text{Exec}_\Theta(z; G)$ of a latent biosynthetic program z executed under an operator library Θ and a domain-derived grammar G ; genome mining is inference over z from observables, and the field’s ceiling is a statement about grammar *identifiability* rather than data volume (Eilertsen, 2026). That framework is the design paper’s subject; here we turn the same machinery on a different question. The executor is *sound*—it builds only mass- and chemistry-consistent structures, and the verifier rejects the rest—so its set of reachable products is a precise, certifiable object. We ask: over the MIBiG-deposited fungal PKS manifold, where does that reach stop, and what *kind* of hardness is each gap?

This is a negative-space characterization, and one distinction governs every number below. The executor’s reach defines what is *reachable*; it is silent about what is not. Out-of-grammar chemistry

does not appear in the reachable set as a hard case—it is *absent* from it entirely. So every boundary we report is the boundary of the current sound grammar’s reach, not a claim about where the manifold itself stops, and every fraction is exact-over-reachable and silent-about-unreachable. We say “the executor’s reach,” never “the biological limit,” because the map does not know the latter.

What “sound” and “verified” each guarantee. The two carry deliberately-divided weight, and the division becomes load-bearing once aromatics enter. *Soundness* is the executor’s conservation guarantee together with the verifier’s formula prefilter: every structure built is mass- and chemistry-consistent, and formula-inconsistent candidates are rejected—exactly as strong as stated, no weaker. The VERIFIED verdict is *strictly stronger*: graph-isomorphism ($\simeq$) to the true deposited product, not mere mass-consistency. For most chemistry the two travel together, but for *regiochemically-degenerate* classes—aromatics above all, where ring-oxygen position is formula-invariant—a wrong-regiochemistry product is fully mass-sound yet not $\simeq$ the truth. Mass-soundness alone therefore does not certify regiochemistry; $\simeq$ does, and $\simeq$ ’s correctness for an aromatic core rests on the encoded closure being chemically right—*curated domain chemistry*, not a mechanical guarantee. “Reachable”/VERIFIED means $\simeq$ -correct throughout, and for aromatic cores that correctness inherits the verified closure regiochemistry.

2 Panel 1 — the boundary (certified)

Run inverse across all 326 deposited fungal PKS, exact reconstruction reaches 8/326: the typed verifier returns a non-empty, exactly-matching Z^* for eight cores and a VERIFIED / UNDER-OBSERVED / OUT-OF-GRAMMAR verdict for the rest (Eilertsen, 2026). This 8/326 is the boundary of the current sound grammar’s reach—more precisely, of the sound grammar *plus the curated regiochemistry* of its encoded closures: orsellinic and 6-MSA are reachable because their aromatic regiochemistry was author-verified and encoded, so the eight are $\simeq$ -exact, inheriting that curated correctness rather than pure mass-soundness (the honest form of G1). We do not re-derive it here; we take it as the definition of “reachable” that the other two panels stand on, and we carry its two caveats forward unchanged: the corpus is deposited and study-biased (G2), and the count is exact over what the grammar reaches and silent about what it does not (G1).

3 Panel 2 — the frontier (certified)

Within the reachable cores, how much of the per-cycle decision problem is genuinely conformational—resolvable only by a pocket coordinate a synthase-local view cannot supply? Walking the consistent set as a prefix trie and enumerating to exhaustion on the distinct reachable chemistries shows this frontier is thin and *localized*, and exact rather than cap-limited: it is zero over the non-methylating aromatic interiors (label-symmetry is structurally impossible without a methylation firing) and single-digit within the highly-reducing subclass, sharpest in the small, methylation-position-ambiguous clusters (Eilertsen, 2026). We import the result as the second face of the map and note only its governing caveat (G1): it is exact over *reachable* cores; a larger conformational frontier can hide behind the coverage boundary that Panel 3 measures, among the cores the grammar cannot yet build.

class	cores	kind of hardness
reachable (exact)	8	built today
CHO, in-scope	206	add a polyketide tailoring operator (§5)
(confirmed unreachable	203	)
(inconclusive, budget	3	reachability not exhaustively confirmed)
heteroatom, scope boundary	112	a different grammar (NRPS-hybrid / halogenase)
total	326	MIBiG-deposited, study-biased

Table 1: The coverage partition (certified). Counts are exact over the executor’s reach (G1); the corpus is the MIBiG-deposited manifold (G2).

4 Panel 3 — the coverage census

The new work of this paper is to characterize the 318 non-reachable cores: not merely to count them, but to sort them by *what kind* of operator each is missing, and to size how concentrated that demand is.

The partition (certified). Table 1 gives it. Of 326 deposited cores, 8 are reachable; 206 are CHO chemistry the polyketide grammar could in principle reach but does not (the *in-scope* gap); 112 contain non-CHO atoms (N, halogen, S) and so need machinery outside the polyketide reduction grammar entirely (the *scope boundary*). These are different questions and we never blend them.

The boundary is genuine, not a scan artifact. The reachability scan caps exact-match search at $C \leq 20$ for tractability, so 94 large cores were initially untested. We re-scanned all 94 with the cap lifted to $C80$. *Zero* of the 91 conclusively tested large cores were reachable (the remaining 3 exhausted the reduced search budget and are held as inconclusive, not folded into “unreachable”). The size cap was concealing no reachable cores: the out-of-grammar counts are genuine gaps in the grammar’s reach, not a tractability limit. This forecloses the most immediate objection to Panel 1—that the boundary is an artifact of where we stopped scanning—and we state it as the result it is.

Where the in-scope hardness lives (roadmap; structural-proxy upper bound). We tag each in-scope core by the coarse operator class(es) its structure requires—*aromatic_cyclization*, *oxidative_tailoring*, *non_aromatic_ring/rearrangement*, *prenylation*, and two catch-alls—anchoring the oxidation tag to the grammar’s formula budget (a core that becomes formula-feasible after removing k oxygens has an oxidation gap, not a topology gap). A greedy set-cover over these classes (Table 2) is steep: a few coarse families proxy-cover the bulk. **This curve is a structural-feature proxy and an explicit upper bound on recovery:** a core counts as proxy-recovered when all its tagged classes are present, which is necessary but not sufficient—the tag does not guarantee the operator can be written *soundly*, and the non-aromatic families are coarse grab-bags that will fracture into several real operators. The steepness measures how few *chemical families* the demand falls into, not how few operators a solution needs. The dominant family is *non-aromatic ring / macrolactonization*—present in 176 of 206 cores, the resorcylic-acid lactones (zearalenone, hypothemycin) and mycophenolic acid among them—not aromatic cyclization, and it is a hard, soundness-open class.¹

¹The operator-class tagger’s ring detection was corrected to count heterocyclic (lactone) rings, not only all-carbon rings; this reclassified the resorcylic-acid-lactone family from clean-aromatic to aromatic-plus-macrolactonization

#operator classes	+ class added	cumulative proxy-recovered (of 206)
1	non_aromatic_ring/rearrangement	83 (40%)
2	aromatic_cyclization	159 (77%)
3	oxidative_tailoring	187 (91%)
4	prenylation	197 (96%)
5	other_formula	203 (99%)
6	other_topology	206 (100%)

Table 2: In-scope recovery, greedy set-cover. **Structural-proxy UPPER BOUND** (G3/G4), not measured recovery. Concentration into few coarse families is the signal; the per-step counts are not a count of buildable operators.

operator-class set	cores	% of 206
clean aromatic (aromatic cyclization only)	23	11
single hard (rearrangement / oxidation / formula only)	83	40
single soft (topology / prenylation only)	3	1
aromatic + hard (entangled)	85	41
other entangled (two classes)	12	6
total	206	100

Table 3: Canonical mutually-exclusive decomposition of the 206 in-scope cores by operator-class set (roadmap; structural-proxy tags, G4). Hard = {rearrangement, oxidation, formula change}. The 11/86/47% views in the text are overlapping projections of this single partition; the 3 inconclusive cores are included, tagged structurally.

How much of the concentrated part is clean (roadmap; the decisive split). Steepness alone is misleading, because the proxy is loosest on cores that need more than one class. We fix one canonical, mutually-exclusive decomposition of the 206 (Table 3), tagging the *hard*, *possibly-unsound* families as {non-aromatic ring/rearrangement, oxidative tailoring, formula change}; aromatic cyclization is the clean buildable class, and other-topology / prenylation are held as *soft*—meaning only *not the rearrangement/oxidation soundness wall* (a candidate new release mode and a discrete known addition, respectively), with their own sound expressibility *otherwise unassessed*, so they do not silently join the buildable count. The 3 inconclusive cores (Table 1) are included here, tagged by structure despite inconclusive reachability. From this one partition the headline views follow—and they *overlap*; they are projections, not addends: the **clean single-class aromatic** pile is 23 (11%); cores that **touch a hard family** number 178 (86% = 83 single-hard + 85 aromatic-entangled + 10 other-entangled); cores **entangled** across two or more classes number 97 (47% = 85 + 12). So building the aromatic operators soundly recovers *at most* the 23 clean cores; the 85 aromatic-but-entangled need a hard operator too and may not be soundly expressible. That 86% of in-scope cores touch a hard, soundness-open family—not the curve’s steepness—is what governs whether solving the in-scope core is a finite grind or stalls, and it says: mostly it stalls.

The scope boundary (heteroatom). The 112 non-CHO cores are dominated by nitrogen: 98 of the 112 contain N, of which 90 need only an N-incorporation operator and 8 need nitrogen *plus* a

(entangled), reducing the clean single-class count from 76 to 23 and raising the entangled fraction from 19% to 47%. The corrected figures are reported throughout; the correction moved the count in the conservative direction (less buildable, more entangled).

halogen or sulfur operator; halogen appears in 22 and sulfur in 1. These are not a failure of the polyketide grammar; they are the target of a different grammar—NRPS-hybridization and dedicated halogenase/heteroatom machinery—and define the boundary of pure polyketide space. We report them as that boundary and do not re-annex them into “fungal PKS.”

5 The located next build

The census specifies the roadmap, and it specifies it as three different *kinds* of work.

Bounded build: nearly empty. The one coherent, soundly-expressible in-scope family is aromatic cyclization—the class where the mass/formula verifier, plus a regiochemistry-aware harness that gates on graph-isomorphism and verified provenance (§1), can certify a new closure. But the corrected census shows it is almost entirely entangled. Of the 206 in-scope cores only 23 (11%) are clean single-class aromatic, and of those only *one* (olivetic acid) is a pure verified-mode product recoverable by extending an already-encoded closure; the other 22 use PT cyclization modes the executor has not yet encoded. Those modes are finite and sequence-predictable (§6), so closing them is a per-*mode* curation—a handful of classes, each confirmed against the primary literature once, not 22 separate decisions—but each is new encoding, so the safe-extension *measured* ceiling *today* (no new modes added) is ≈ 1 core ($\approx 0.5\%$). The honest floor today is 8 reachable plus 0 clean-buildable-today, and the soundly-extendable gain is a single core. We report it as that, not as a build to schedule: the encouraging “aromatic quarter” of the first pass was a tagging artifact, and the corrected map says even the buildable-looking family is mostly entangled with the hard operators below.

Research bet, and now the dominant pile: rearrangement and oxidation. 86% of in-scope cores touch a hard family, chiefly *macrolactonization* (non-aromatic ring, in 176 of 206), with oxidative tailoring layered on; these route through operators whose *sound expressibility is unproven*. This is most of the in-scope gap, not a remainder. We flag it as an open research question, not as scheduled engineering; the learned, gene-conditioned operator route—templates licensed by a BGC’s tailoring genes, admitted only when they pass the verifier’s mass/formula gate—is the candidate, but whether it covers these families soundly is exactly what is unknown. If a substantial fraction is soundly inexpressible, complete genome-to-product prediction is permanently partial, which is a finding to state.

Unassigned residual: the soft classes. A further 3 in-scope cores (1%; Table 3) carry only a soft class (other-topology or prenylation). These are *not yet assigned* to buildable-versus-research-bet: prenylation is a discrete, plausibly sound addition that may join the build, while other-topology is the grab-bag that fractures in Phase 1 and could resolve either way. The three-way roadmap covers the rest; the residual is named here rather than folded into the buildable count, pending a per-class soundness assessment.

Different grammar: the heteroatom third. The scope boundary (§4) is handed to NRPS-hybrid machinery, not addressed here.

6 The instrument: a decision procedure for sound expressibility

The map locates where the hardness is; the durable object is a procedure *for* adjudicating each piece—*proposed* here, and demonstrated on the one class the map has actually resolved. An operator O is **soundly expressible from φ iff the formula-invariant degree of freedom it must resolve is a computable function of the recoverable features $\varphi(B)$** . Soundness (mass) is free by construction; $\simeq$ -correctness is the bite; the DOF is exactly where $\simeq$ and mass diverge—ring-oxygen position for aromatics, ring size for macrolactones. The procedure places an operator on a cost ladder, beneath which sits one prior question, and it reports a verdict only where the evidence reaches one.

Rung 0 — the φ -scope prior. Does the determinant even live in the unit φ ranges over? $\varphi(B)$ is the cluster. If the deciding enzyme is *trans-acting*—a separate gene that need not be co-clustered—the determinant is outside cluster- φ , a scope failure no per-cluster operator can repair. This is asked first because it is cheap, corpus-answerable from annotations, and *free of deposition bias*: “is there an annotated releasing domain in this cluster” is a structural fact about the deposit, not a chemistry outcome enriched by what gets published.

The cost ladder. Fix φ once as *the cluster*—the BGC—since the stated problem is BGC→product; “recoverable” means recoverable from the cluster, and that fixing is also what makes rung 0 well-posed. With φ_{chain} the upstream-determined linear precursor and the cluster-encoded releasing enzyme contributing its sequence, fold, and dynamics in turn: (1) $f(\varphi_{\text{chain}})$ — trivial, the chain fixes it; (2) $f(\varphi_{\text{chain}}, \text{enzyme sequence})$ — *curatable from sequence* (cheap; a finite rule set keyed to a sequence class, the PT-clade pattern); (3) $f(\varphi_{\text{chain}}, \text{enzyme fold/dynamics})$ but not the sequence directly — *recoverable through the fold*: AF2 maps sequence→fold and, in principle, fold→kinetics by MD/QM, so even kinetic or dynamic selection is a function of the cluster at *escalating, possibly ruinous* cost—a costly rung 3, not an impossibility; (4) *not a function of any cluster feature* $\Rightarrow$ *soundly inexpressible*: no sound *and* complete grammar exists for the class. Exactly two conditions reach rung 4: (i) the outcome is intrinsically a *distribution*—the enzyme makes a genuine mixture, so there is no single product and no function maps to a non-point (a within-enzyme collision is its signature); (ii) the determinant lies *outside the organism’s genome* (a host or environmental input). A trans-acting enzyme encoded *elsewhere in the genome* is not rung 4 but a rung-0 scope failure, and kinetic/dynamic selection is a costly rung 3—so the impossibility bar is high and narrow, consistent with the honest ceiling being “expressible at escalating cost almost everywhere.” Discriminators: rung 0 by cis-vs-trans annotation; rung 4(i) by a within-enzyme collision (one sequence, one substrate, ≥ 2 products); rung 2-vs-3 by a structure-vs-sequence experiment with φ_{chain} regressed out. Each discriminator’s false reads are catalogued—deposition bias, phylogenetic confounding, base-rate inflation, low-pLDDT folds, sequence-proxy circularity, static-fold blindness—because every one of them manufactures the encouraging answer.

Resolved exemplar: aromatic cyclization (rung 2). The PT domain is integral to the NR-PKS (rung 0 passes, cis), and—decisively for rung 2—PT cyclization regiochemistry is a *finite, sequence-predictable* map: fungal PT domains fall into a handful of phylogenetic classes whose regioselectivity (C2–C7, C4–C9, C6–C11) has been classified *and prospectively verified* from sequence (Fungal PT regioselectivity, 2010). So the formula-invariant DOF is a function of the enzyme’s sequence—a finite rule set, not a per-core lookup. The executor encodes those modes as closures validated under $\simeq$, and “literature confirmation” is a per-*mode* check of the sequence-keyed

rule (a handful of modes), which is what places orsellinic and 6-MSA among the reachable 8 (§2). Aromatic cyclization is therefore a *resolved* rung-2 instance—the instrument returning a predictive verdict, not buildability once handed the answer.

Located, not yet resolved: macrolactonization. Rung 0: for the dominant resorcylic-acid-lactone family the releasing thioesterase is a *cis* C-terminal NR-PKS domain (studied in vitro as an excised TE), so the determinant is in-cluster and rung 0 passes; the trans-acting minority (e.g. certain polyene macrolides) is the rung-0-failure subset, flaggable from annotations. Rung-4(i)—the intrinsic-distribution route—is closed: the characterized zearealenone TE is regiospecific on a fixed substrate (one substrate, one ring), so the outcome is a point, not a mixture. So macrolactonization sits on rungs 2–3—curatable at a cost not yet measured—and the structure-vs-sequence experiment is the live discriminator. The instrument has thus *located* macrolactonization (not rung 0, not rung-4-stochastic) without resolving its cost, and that pending state is reported as pending, not as a verdict.

7 Contribution

A negative-space characterization is a contribution because it converts “fungal PKS biosynthesis is hard” into an *actionable partition*: it tells every group which fraction is a bounded engineering build, which is an open research question, and which belongs to a different enzyme family—where to spend effort and what kind of effort each piece needs—which a partial solver does not. Beyond the partition, the durable object is the decision procedure (§6): a single criterion—an operator is soundly expressible iff its formula-invariant DOF is a function of recoverable φ —with discriminators that place each operator class on a cost ladder, demonstrated *resolved* on aromatic cyclization (rung 2, the 8) and *located* on macrolactonization (rung 0 passed, the stochastic route to inexpressibility closed, cost still open). We offer it as a *proposed* procedure—a criterion, one worked verdict (aromatics), and one application in progress (macrolactonization)—not a general adjudicator: its reach across the remaining classes is a claim to be earned one class at a time, which this paper does not yet make. The map is complete as a characterization and explicit about where it stops: the boundary and frontier are certified (exact, exhaustion-checked, over reachable cores), the partition is certified, the recovery curve is a stated upper bound, and the soundly-extendable part is measured at $\approx$ one core—the rest gated on confirmed regiochemistry and on operators whose soundness is unproven. Its value is not a prediction accuracy; it is the located statement of what fungal-PKS prediction *requires*, drawn by a tool sound enough to make “reachable” a certifiable word.

References

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